Retail Sales Forecasting

Predicting Future Sales from 540K+ Historical Retail Transactions Using Python & Machine Learning

540K+	19.84%	5 Models
Transactions	Test MAPE	Compared

github.com/kemasmalfath/sales-forecasting-portfolio-project

Data Cleaning

Business-Driven Preprocessing Rules

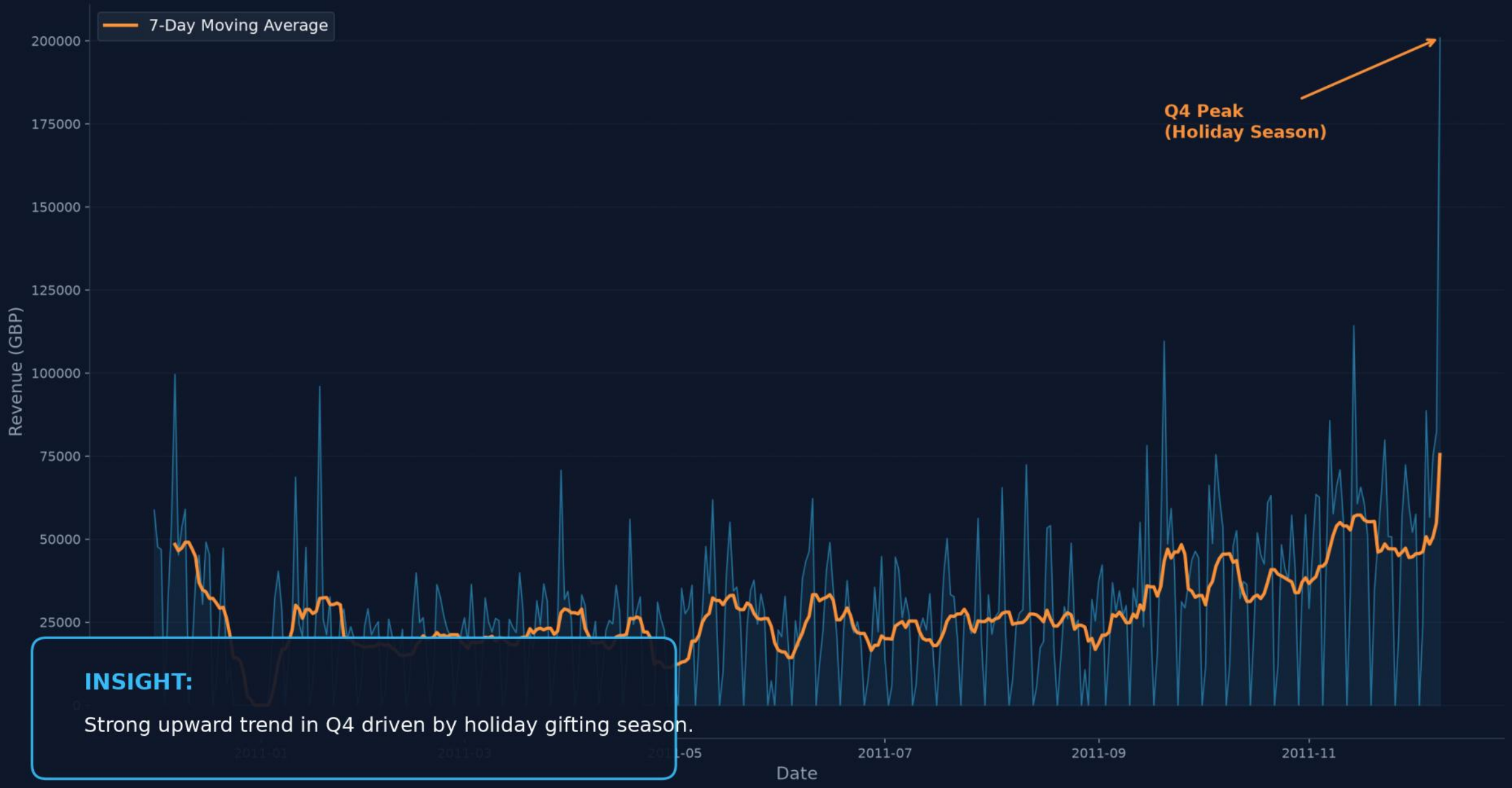
Raw Dataset	**541,909**
After Deduplication	**536,641**
After Removing Cancellations	**527,390**
After Filtering Price <= 0	**524,878**

KEY DECISION:

Preserved 132K rows with missing CustomerID (~25% of all transactions).

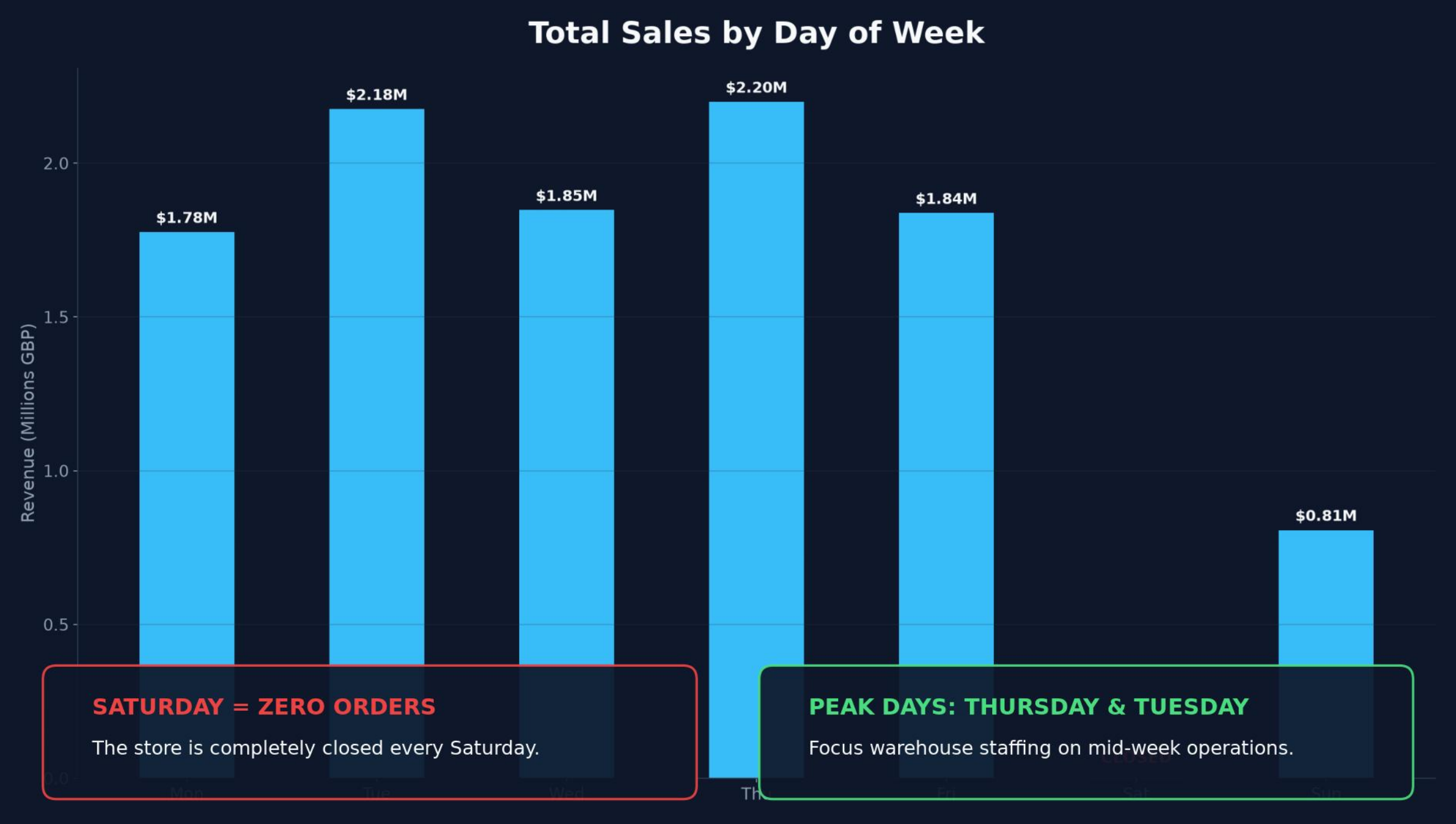
Guest shoppers still generate real revenue. Dropping them = 25% demand underestimation.

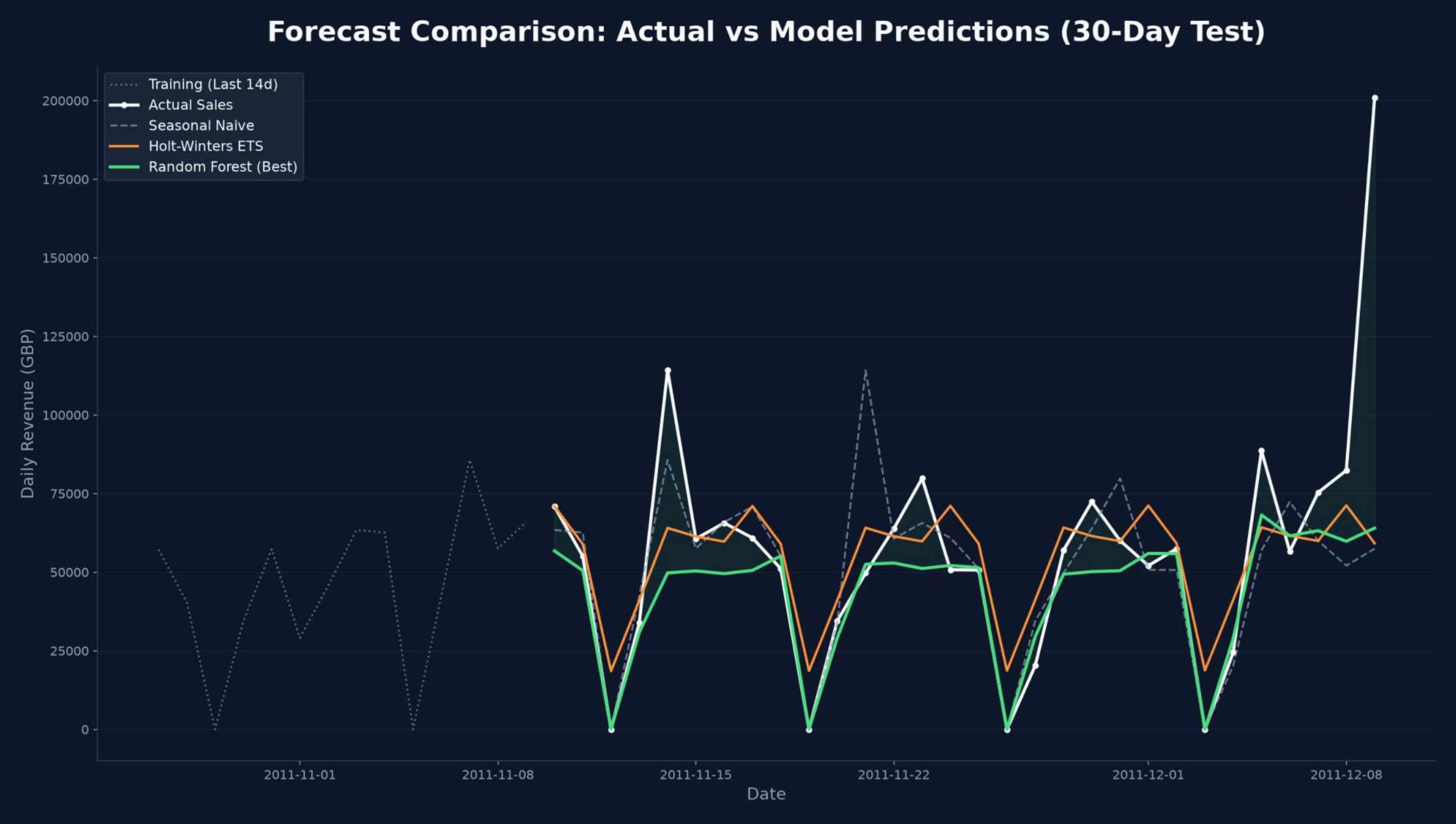
Daily Sales Trend & 7-Day Moving Average



INSIGHT:

Strong upward trend in Q4 driven by holiday gifting season.


Total Sales by Day of Week
Revenue (Millions GBP)
2.0
1.5
1.0
0.5
0.0
$1.78M
$2.18M
$1.85M
$2.20M
$1.84M
$0.81M
Mon
Tue
Wed
Th
Fri
Sat
Sun
SATURDAY = ZERO ORDERS
The store is completely closed every Saturday.
PEAK DAYS: THURSDAY & TUESDAY
Focus warehouse staffing on mid-week operations.


Forecast Comparison: Actual vs Model Predictions (30-Day Test)
Training (Last 14d)
Actual Sales
Seasonal Naive
Holt-Winters ETS
Random Forest (Best)
Daily Revenue (GBP)
200000
175000
150000
125000
100000
75000
50000
25000
0
2011-11-01
2011-11-08
2011-11-15
2011-11-22
2011-12-01
2011-12-08
Date

Model Scoreboard

30-Day Out-of-Sample Test Evaluation

Model	MAE	RMSE	MAPE
Random Forest (ML) BEST	**$14,414**	**$29,697**	**19.8%**
Seasonal Naive (7-Day)	$15,372	$31,289	24.3%
Holt-Winters ETS	$16,896	$30,350	24.1%
Standard Naive	$29,825	$39,824	42.8%

WHY RANDOM FOREST WON:

Combined multi-period lag momentum (Lag 1, 2) with weekly cyclical seasonality (Lag 7, 14) and calendar signals.

Business Impact

Actionable Recommendations from Forecasting Results

Inventory Safety Buffer

15% - 20%

safety stock buffer on fast-moving products during Q4 peak season

Staffing Optimization

Tue - Thu Focus

Schedule peak warehouse shifts mid-week. Zero staff needed on Saturdays.

Limitations & Next Steps

1-year data only: yearly seasonality patterns cannot yet be verified.

No exogenous features: promotions, ad spend, and weather data would improve accuracy.

Full reproducible pipeline available on GitHub with Google Colab support.